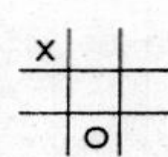
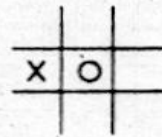
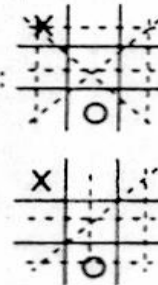


Applying MiniMax to tic-tac-toe

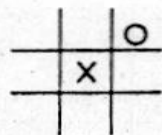
- The static evaluation function heuristic



X has 6 possible win paths:
O has 5 possible wins:
 $E(n) = 6 - 5 = 1$



X has 4 possible win paths;
O has 6 possible wins
 $E(n) = 4 - 6 = -2$



X has 5 possible win paths;
O has 4 possible wins
 $E(n) = 5 - 4 = 1$

Heuristic is $E(n) = M(n) - O(n)$

where $M(n)$ is the total of My possible winning lines

$O(n)$ is total of Opponent's possible winning lines

$E(n)$ is the total Evaluation for state n

Figure 4.16 Heuristic measuring conflict applied to states of tic-tac-toe.

Backup Values

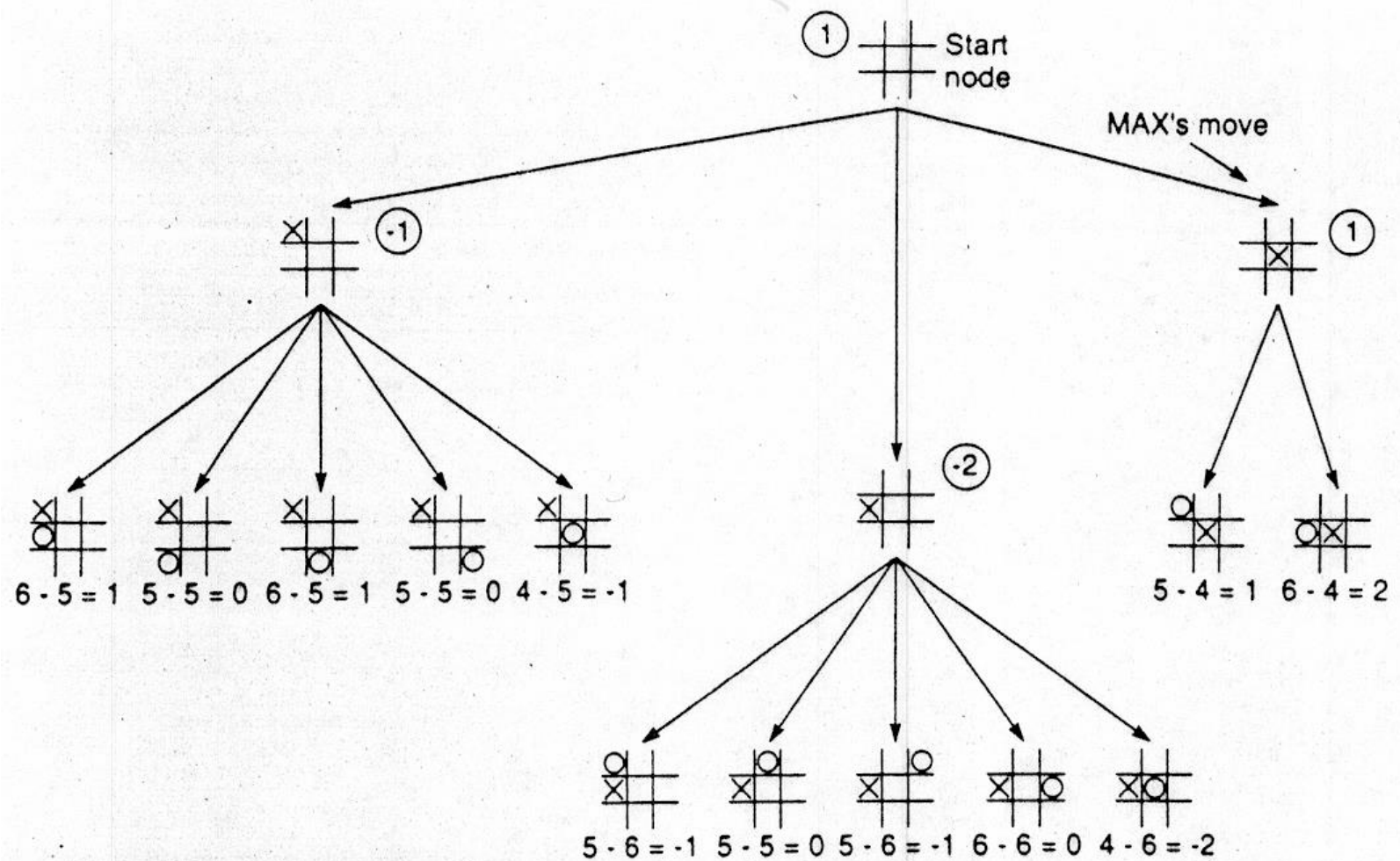


Figure 4.17 Two-ply minimax applied to the opening move of tic-tac-toe.

Tic-Tac-Toe Example with Alpha-Beta Pruning

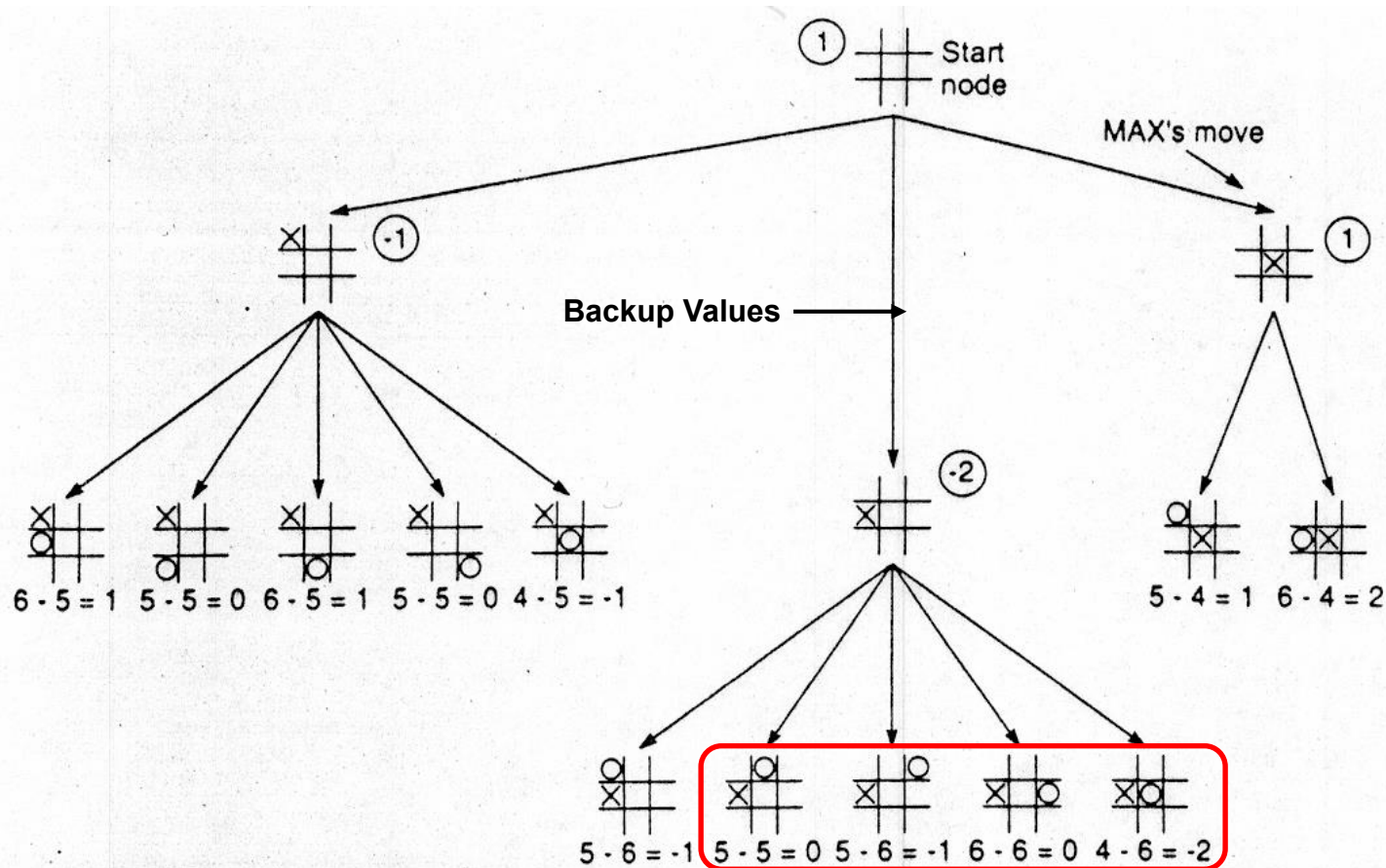


Figure 4.17 Two-ply minimax applied to the opening move of tic-tac-toe.